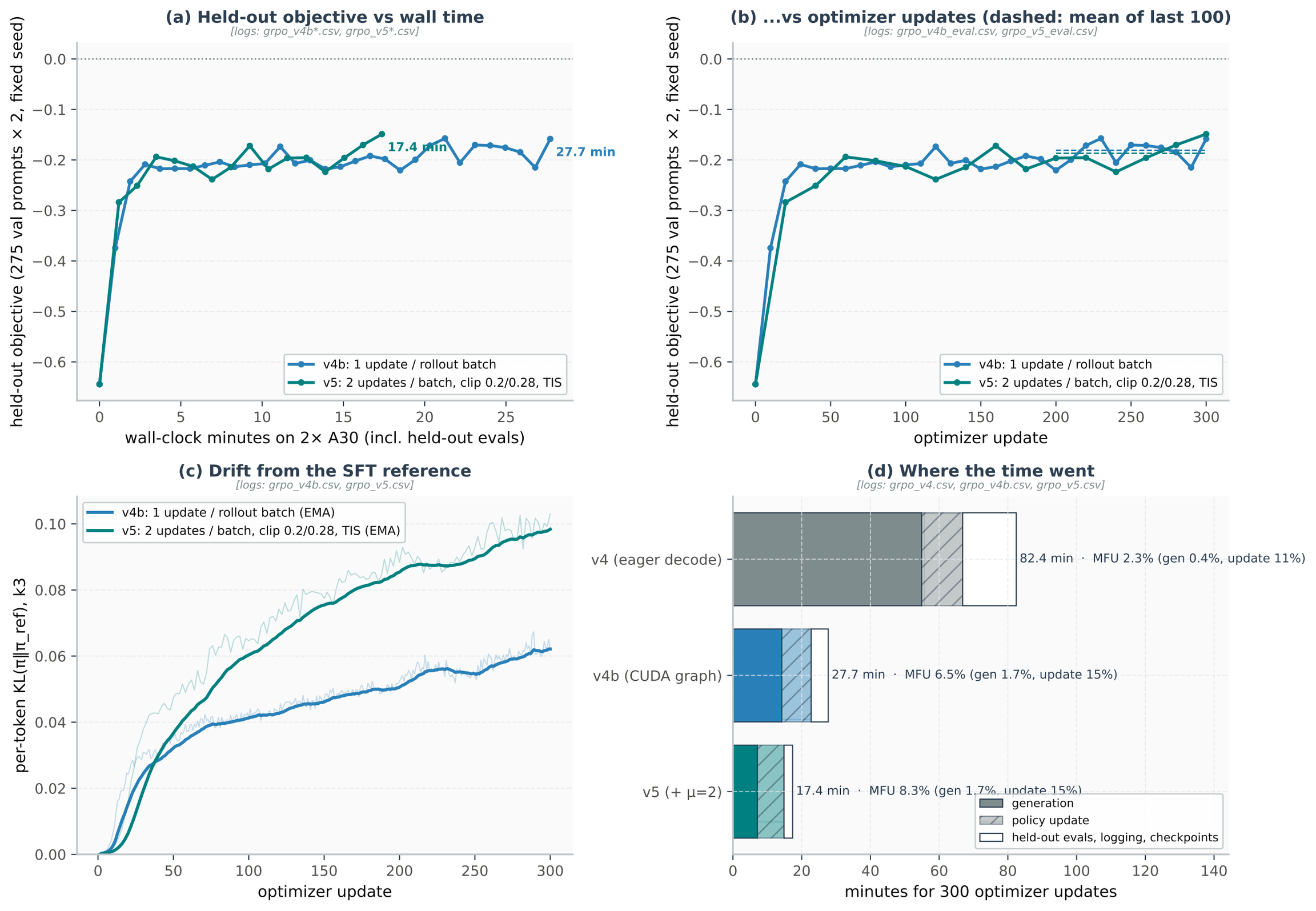

(a) Held-out objective vs wall time
[logs: grpo_v4b*.csv, grpo_v5*.csv]
held-out objective (275 val prompts × 2, fixed seed)
wall-clock minutes on 2× A30 (incl. held-out evals)
17.4 min
27.7 min
v4b: 1 update / rollout batch
v5: 2 updates / batch, clip 0.2/0.28, TIS
(b) ...vs optimizer updates (dashed: mean of last 100)
[logs: grpo_v4b_eval.csv, grpo_v5_eval.csv]
optimizer update
(c) Drift from the SFT reference
[logs: grpo_v4b.csv, grpo_v5.csv]
per-token KL(π‖π_ref), k3
v4b: 1 update / rollout batch (EMA)
v5: 2 updates / batch, clip 0.2/0.28, TIS (EMA)
(d) Where the time went
[logs: grpo_v4.csv, grpo_v4b.csv, grpo_v5.csv]
v4 (eager decode)
v4b (CUDA graph)
v5 (+ μ=2)
82.4 min · MFU 2.3% (gen 0.4%, update 11%)
27.7 min · MFU 6.5% (gen 1.7%, update 15%)
17.4 min · MFU 8.3% (gen 1.7%, update 15%)
generation
policy update
held-out evals, logging, checkpoints
minutes for 300 optimizer updates